

DIFFERENT CLUSTER SETS RESULT IN SIMILAR METABOLIC, ENDOCRINE, AND PERCEPTUAL RESPONSES IN TRAINED MEN

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ABSTRACT

Tufano, JJ, Conlon, JA, Nimphius, S, Oliver, JM, Kreutzer, A, and Haff, GG. Different cluster sets result in similar metabolic, endocrine, and perceptual responses in trained men. *J Strength Cond Res* 33(2): 346–354, 2019—The purpose of this study was to compare the kinematic, metabolic, endocrine, and perceptual responses of 3 back squat protocols with equal loads, number of repetitions, and total rest duration. Eight strength-trained men performed 36 back squats using 75% 1 repetition maximum and 420 seconds of total rest during basic cluster sets of 4 (CS4), rest redistribution sets of 4 (RR4), and rest redistribution sets of 1 (RR1). Rating of perceived exertion (RPE), blood lactate (La), mean velocity maintenance (MVM), and mean velocity loss (MVL) were measured during exercise. Total testosterone (TT), growth hormone (GH), cortisol (C), and sex hormone-binding globulin (SHBG) were measured before exercise and 15, 30, and 60 minutes post-exercise. There were no differences between protocols for MVM. However, MVL was less during RR1 compared with RR4 ($p = 0.032$), and neither protocol was different than CS4. All protocols resulted in similar increases in RPE and La, which remained elevated up to 30 minutes postexercise ($p \leq 0.05$). In all protocols, GH increased and returned to baseline by 60 minutes postexercise ($p \leq 0.05$). At 60 minutes post-exercise, TT was less than all other time points ($p \leq 0.05$). There were no main effects for time for SHBG or C. The data from this study show that different types of cluster set protocols can result in proanabolic physiological responses to resistance training. In addition, coaches can redistribute rest periods without affecting perceived effort or metabolic and

hormonal changes if the external load, number of repetitions, and total rest time are equalized.

KEY WORDS intraset rest, strength, hypertrophy, power, squats, fatigue

INTRODUCTION

Resistance training performed with traditional sets (TS), in which lifters consecutively perform repetitions within a set, is fatiguing and results in decreased energy availability (9,10), increased metabolic stress (9,33), and decreased movement velocity (38,45). For example, when performing 3 sets of 12 repetitions with moderate-to-heavy loads and 2 minutes of interset rest intervals, movement velocity, and power output decrease throughout each set, as energy availability decreases and fatigue ensues (33,38,45). To combat such fatigue-induced decreases in acute performance, researchers have begun to investigate the effects of intraset rest periods in set structures known as cluster sets (CS) (43). There are many different ways in which CS can be made, including the addition of intraset rest periods, known as a basic CS (i.e., 3 sets of 12 with 2 minutes of interset rest periods plus additional rest periods of 15–30 seconds dispersed within each set), and the redistribution of rest (RR) in which total rest time is divided to create shorter but more frequent rest periods (i.e., 3 sets of 12 with 2 minutes of interset rest divided into 6 sets of 6 with 1 minute of interset rest) (43). Although these 2 subtypes of CS have been investigated independently (7,15,20,30,34,43–45), to the authors' knowledge, they have never been compared with each other within the same study. Nonetheless, it has been hypothesized that the use of all types of CS allow for the frequent and partial replenishment of adenosine triphosphate (ATP) and phosphocreatine (PCr) stores, resulting in a better ability to maintain velocity and power output compared with TS (15,43).

Previous research has shown that various types of CS performed with different exercises and intensities generally

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33(2)/346–354

Journal of Strength and Conditioning Research
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result in better acute performance maintenance (i.e., faster movement velocity, greater power output, greater jump height) compared with TS (7,20,30,34,43). In addition to the positive effects of CS on acute performance, research has indicated that implementing less-fatiguing protocols during training can result in similar strength gains compared with more fatiguing protocols (6,18,19). Although these studies indicate that large amounts of fatigue may not be necessary to induce gains in strength, it should not be forgotten that some degree of acute fatigue is needed to continually stress the body and stimulate neuromuscular adaptations (41). Furthermore, fatigue is often warranted when seeking to increase skeletal muscle hypertrophy (24,35,40).

To acutely increase fatigue and stimulate the anabolic process, classic body-building programs include moderately heavy loads performed for many repetitions with short rest periods (3,5,39). Historical works from various laboratories have shown that such body-building regimes inclusive of high training volumes and short rest periods result in large amounts of external work, increased metabolic stress, and changes in acute hormonal concentrations (13,21,22), resulting in associations to be made between these variables (24,36,42). Despite a plethora of data showing that high-volume resistance training increases the acute endocrine responses during and after exercise and that such responses play a role in anabolism, some researchers have begun to question the importance of these responses for inducing skeletal muscle hypertrophy (46). Nevertheless, previous evidence relating acute increases in hormone concentrations to skeletal muscle growth and tissue regeneration should not be disregarded (24), and new theories open the door for unconventional training methods that use a variety of loads and rest period frequencies and durations (1,18,47).

The ability to adjust a seemingly infinite number of external loads, rest periods, and training volumes allows for novel training stimuli to be introduced at various times within a training program. Specifically, additional rest periods provided during CS can allow for greater training loads and a greater amount of total work compared with a TS protocol of equal relative fatigue (44). Such increases in mechanical stress could possibly affect acute metabolic demands during exercise (40), postexercise endocrine responses (27,29), and skeletal muscle growth (25).

To date, a few studies have investigated the effect of CS on metabolic and endocrine responses; but, these studies have used divergent protocols, making direct comparisons between studies difficult (7,12,33). For example, Oliver et al. (33) employed an RR protocol during 4 sets of 10 back squats using 70% of 1 repetition maximum (1RM), whereas Girman et al. (7) implemented an RR protocol during upper- and lower-body circuit training using a wide variety of exercises and training loads. Although Girman et al. (7) and Oliver et al. (33) used RR protocols, Goto et al. (12) implemented a basic CS protocol inclusive of 30 seconds of intra-set rest periods during 3–5 sets of machine exercises with

a 10 repetition maximum (10RM) load. Despite a lack of uniformity among the protocols, all the sessions increased blood lactate (La), but the response was always greater in response to TS (7,12,33). In addition, RR and TS resulted in increased growth hormone (GH) concentrations after exercise (7,33), whereas basic CS inclusive of intraset rest periods did not (12).

Together, the data from these 3 studies (7,12,33) indicate that RR and basic CS protocols both have the ability to increase La; but, RR may also increase acute endocrine responses, whereas basic CS may not. In addition, none of the studies reported the differences in subjects' rating of perceived exertion (RPE) between protocols, which may serve as a simple method of subjectively reporting perceptual exertion levels that relate to physiological fatigue (17,28,31). Compared with TS, previous research has shown that RR (28) and basic CS protocols (17) result in lower RPE, and it has been hypothesized that these set structures may also result in less La and a lesser endocrine response (14). However, to the authors' knowledge, La, RPE, and endocrine responses have not been collected during the same study using different types of CS protocols. As CS and RR protocols have already been independently compared with TS but not with each other, the purpose of this study was to compare the effects of a basic hypertrophy-oriented CS protocol (44) with 2 different RR protocols on changes in movement velocity in addition to the hormonal, metabolic, and perceptual responses of back squats in trained men. It was hypothesized that the RR protocol with the most frequent rest periods would result in less La and blunted hormonal responses compared with when rest periods were less frequent.

METHODS

Experimental Approach to the Problem

Using a repeated measures design, subjects reported to the laboratory 4 times. Before the 3 randomized experimental sessions, subjects completed a 1RM session, immediately after which they were familiarized with the procedures of the subsequent experimental testing sessions. Two to 3 days after the 1RM session and after 48 hours of no physical activity, the first experimental session took place. Each of the experimental sessions included 36 back squat repetitions using a load of 75% of 1RM. The total assigned rest time was equal between protocols, but the distribution of rest varied (16,33). The cluster sets of 4 (CS4) protocol included 30 seconds of rest after the fourth, eighth, 16th, 20th, 28th, and 32nd repetition in addition to 120 seconds of rest after the 12th and 24th repetition (44). For the other 2 protocols, the total rest time was redistributed to include 9 sets of 4 repetitions (RR4) with 52.5 seconds of rest provided after every fourth repetition, or 36 sets of 1 repetition (RR1) with 12 seconds of rest provided after every repetition. Therefore, all 3 protocols included 36 repetitions at 75% 1RM with 420 seconds of standing, unloaded rest (Figure 1). Subjects were instructed to avoid all other forms of exercise during the

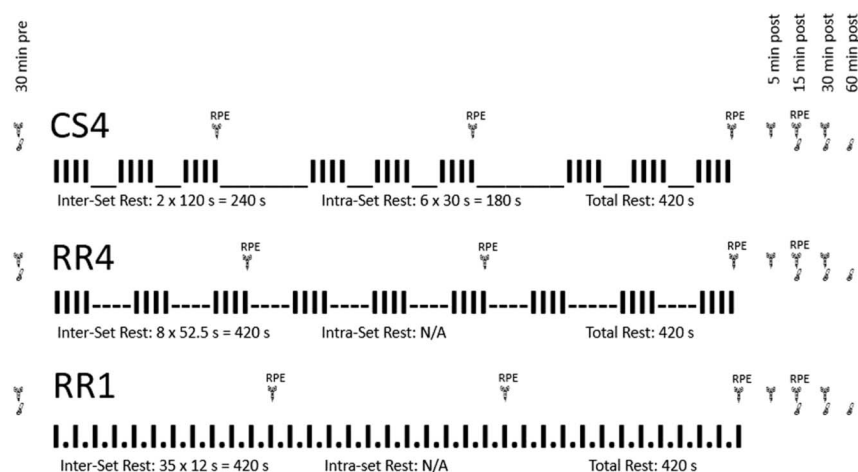


Figure 1. Protocol design for cluster sets of 4 (CS4), rest redistribution 4 (RR4), and rest redistribution 1 (RR1). Time-sensitive data collection included blood lactate (||); Vacutainer blood collection for growth hormone, cortisol, total testosterone, and sex hormone-binding globulin (U); and rating of perceived exertion (RPE). Intra-set rest interval is not applicable (N/A).

study period and abstained from eating and drinking during the protocols. All subjects completed all 36 repetitions in each protocol.

The decision to investigate the CS4 protocol, which served as the “reference” protocol from which the RR1 and RR4 protocols were created, was based on previous data demonstrating that this specific protocol was able to increase mechanical stress compared with a TS protocol without inducing greater neuromuscular fatigue, assessed by relative decreases in movement velocity (44). Because the duration and frequency of rest periods play a large role in the acute metabolic and hormonal responses to exercise, the rest periods from CS4 were redistributed to create the RR1 and RR4 protocols. By assigning rest periods after every 1 or 4 repetitions, the protocol designs in this study represent much of the CS and RR literature (43), allowing for comparisons to be made between studies.

Subjects

Eight resistance-trained men (mean $\pm$ SD: 25.2 $\pm$ 4.1 years, range 18–30 years; 76.7 $\pm$ 5.1 kg; 1.75 $\pm$ 0.07 m) with at least 6 months of strength training experience using the full back squat exercise participated in this study. All subjects could perform a free weight back squat (top of the thighs below parallel) with at least 150% of their body mass and were excluded if they reported any recent musculoskeletal injuries or recent use of ergogenic aids that may have affected endocrine function. Specifically, subjects must not have had any history of anabolic steroid use and could not have used any other supplements that are marketed to influence hormone or insulin levels. The average 1RM was 135.0 $\pm$ 16.8 kg, 1RM to body mass ratio was 1.76 $\pm$ 0.22, and peak knee flexion angle at the bottom of the squat was 129.5 $\pm$ 11.5°.

All procedures were approved by the University Human Research Ethics Committee at Edith Cowan University, and all subjects gave written informed consent before participation.

Procedures

One Repetition Maximum Testing. Subjects reported to the laboratory at the same time of day as the upcoming experimental sessions (in the morning between 06:00 and 10:00). After anthropometric measures, subjects completed a 5-minute stationary cycle warm-up, pedaling at 60 revolutions per minute with 100 W of resistance. Subjects then completed a 5-minute dynamic warm-up consisting of bodyweight squats, leg swings, and various marching exercises followed by 8, 5, and 3 back squat repetitions with 25, 50, and 60% of estimated 1RM. The peak knee flexion angle was measured via handheld goniometry during the final warm-up repetition with the 50% load. During this repetition, subjects paused momentarily at the bottom of the squat, allowing the researchers to measure knee angle and set up the safety bars within the squat rack for the upcoming 1RM attempts. Next, subjects began free weight back squat 1RM attempts starting at 85% estimated 1RM, resting 2–5 minutes between efforts of progressively increasing loads. Each subject’s heel and toe locations were recorded using a horizontal-vertical grid intersecting every 1 cm and were kept constant for all remaining sessions.

Experimental Sessions. After refraining from physical activities outside of normal daily tasks for 48–72 hours, subjects reported to the laboratory in the morning after a 12-hour fast. Subjects were then seated for 15 minutes before pre-exercise blood samples (PRE) were obtained. The warm-up protocol was identical to the 1RM session, and after 5 minutes of rest,

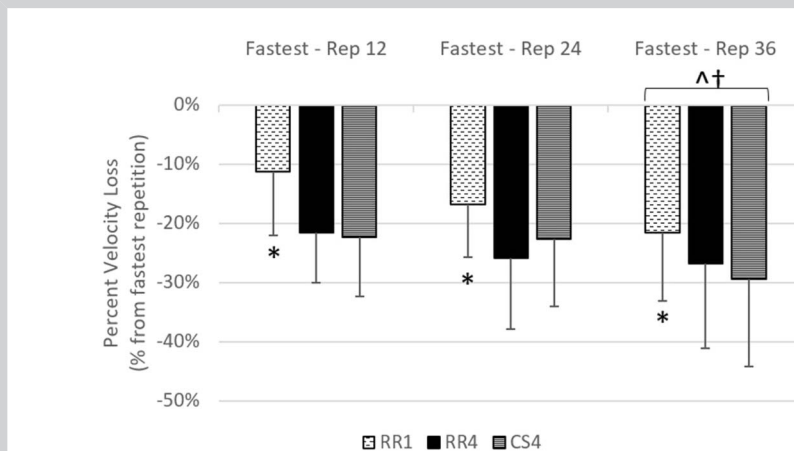


Figure 2. Percent of mean velocity loss (MVL) from the fastest repetition to repetition 12 (Rep12), repetition 24 (Rep24), and repetition 36 (Rep36) during the rest redistribution 1 (RR1), rest redistribution 4 (RR4), and cluster sets of 4 (CS4) protocol. When collapsed across time, MVL was significantly less in RR1 compared with RR4 ($p = 0.032$)*. When collapsed across protocols, MVL from the fastest repetition to Rep36 was significantly greater than the MVL from the fastest to Rep12 ($p = 0.020$)^ and the fastest to Rep24 ($p = 0.011$)†.

subjects positioned themselves under the barbell. After a verbal 5-second countdown, subjects unracked the bar from the squat rack when the count reached “zero.” Once the subjects’ feet were in line with their individual heel and toe locations, subjects performed the desired number of repetitions (4 repetitions in CS4 and RR4, and 1 repetition during RR1). They were instructed to perform the eccentric portion under control and “explode out of the bottom,” returning to a standing position as quickly as possible without jumping or the bar leaving the shoulders (45). After completing the prescribed number of repetitions, subjects reracked the bar and remained standing, unloaded during the rest period. When

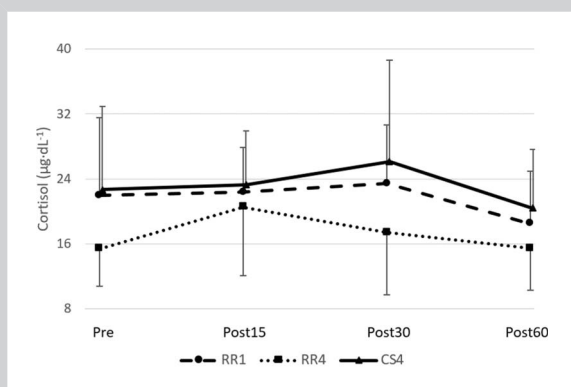


Figure 3. Cortisol before (Pre), 15 (Post15), 30 (Post30), and 60 (Post60) minutes after exercise. When collapsed across all time points, rest redistribution 4 (RR4) was significantly less than rest redistribution 1 (RR1) and cluster sets of 4 (CS4) ($p \leq 0.05$). There were no significant interactions or differences between time points.

the next 5-second countdown began, they positioned themselves under the barbell, ready to repeat the process when the countdown reached “zero.” This was repeated until the final repetition was completed. Upon completion of the final repetition, subjects were seated and remained quietly seated for 60 minutes. Postexercise blood draws were obtained 15 (P15), 30 (P30), and 60 (P60) min after the final repetition was completed. All experimental sessions were separated by 48–96 hours and occurred at the same time of day within a 2-hour time frame.

Kinematic Data Collection and Analysis.

All data were collected using a customized LabVIEW program (National Instruments, Version14.0) and were manually analyzed after data collection. Four linear position transducers (Celesco PT5A-250; Celesco, Chatsworth, CA, USA) were positioned above the squat rack at all 4 corners, sending raw voltage data via a BNC-2090 interface box with an analog-to-digital card (NI-6014; National Instruments, Austin TX, USA). All signals were sampled at 1,000 Hz and filtered using a fourth-order low-pass Butterworth filter with a cut-off frequency of 50 Hz.

Using time and displacement from the transducers, mean velocity was calculated for all back squat repetitions. From these values, mean velocity maintenance (MVM) (45) was calculated to show the average mean velocity of all 36 repetitions in relation to the fastest repetition of each protocol, expressed as a percentage of the fastest repetition. In addition, mean velocity loss (MVL) was calculated by determining the percentage decrease in mean velocity from the fastest repetition (usually the first or second) to the 12th, 24th, and 36th repetition to quantify the degree of neuromuscular fatigue at equal repetitions within each protocol, similar to previous studies (38,45).

Rating of Perceived Exertion. During the 1RM session, subjects were familiarized with a printed 0–10 OMNI-RES scale: a resistance training-specific RPE scale (31). Subjects were then shown the printed scale during the experimental sessions after repetitions 12 (Rep12), 24 (Rep24), and 36 (Rep36) and were asked how difficult the previous repetition was.

Blood Sampling and Analysis. A certified phlebotomist sterilized the area surrounding the medial cubital vein, according

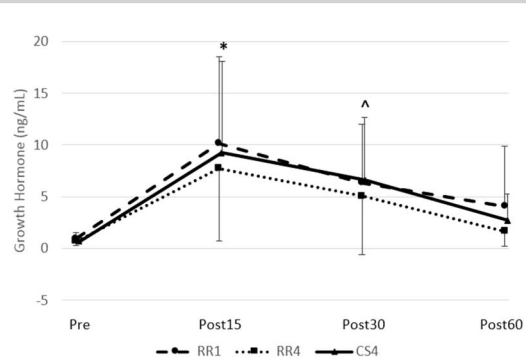


Figure 4. Growth hormone before (Pre), 15 (Post15), 30 (Post 30), and 60 (Post60) minutes after exercise during cluster sets of 4 (CS4), rest redistribution 1 (RR1), and rest redistribution 4 (RR4). Post15 was significantly greater than all other time points ($p \leq 0.05$)*, and Post30 was significantly greater than pre ($p \leq 0.05$)^.

to standard procedures and collected a 5-ml blood sample in a Vacutainer tube containing no additives (BD Biosciences, San Jose, CA, USA) for the analysis of serum total testosterone (TT), cortisol (C), sex hormone-binding globulin (SHBG), and GH. After gently inverting the tube with no additive 5 times, the samples remained at room temperature for approximately 45 minutes and were then stored in a refrigerator for up to 3 hours. The same blood sampling procedures were repeated at P15, P30, and P60. All samples were centrifuged at 3,000 revolutions per minute for 10 minutes in a swing bucket centrifuge. Serum was then

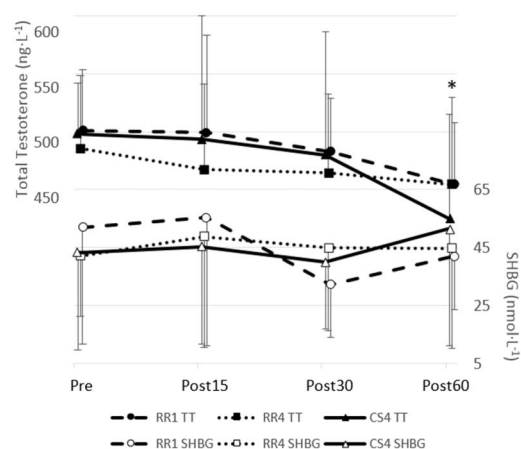


Figure 5. Total testosterone (TT) and sex hormone-binding globulin (SHBG) before (Pre), 15 (Post15), 30 (Post 30), and 60 (Post60) minutes after exercise. When collapsed across protocols, Post60 had significantly less TT than all other time points ($p \leq 0.05$)*. There were no significant differences for SHBG at any time point for any protocol.

aspirated, aliquoted into Eppendorf tubes, and stored at -80°C until analysis.

Using a radioimmunoassay (MP Biomedicals, Orangeburg, NY, USA), all TT and GH samples were run in duplicate on an ISO Data 100 gamma counter (Titertek, Pforzheim, Germany). The interassay coefficient of variation (CV) was 11.88% and 9.74% for TT and GH, respectively. In addition, C and SHBG samples were run in duplicate using an enzyme-linked immunosorbent assay (ALPCO, Salem NH). The interassay CV for C was $<5\%$, and for SHBG, it was 15%.

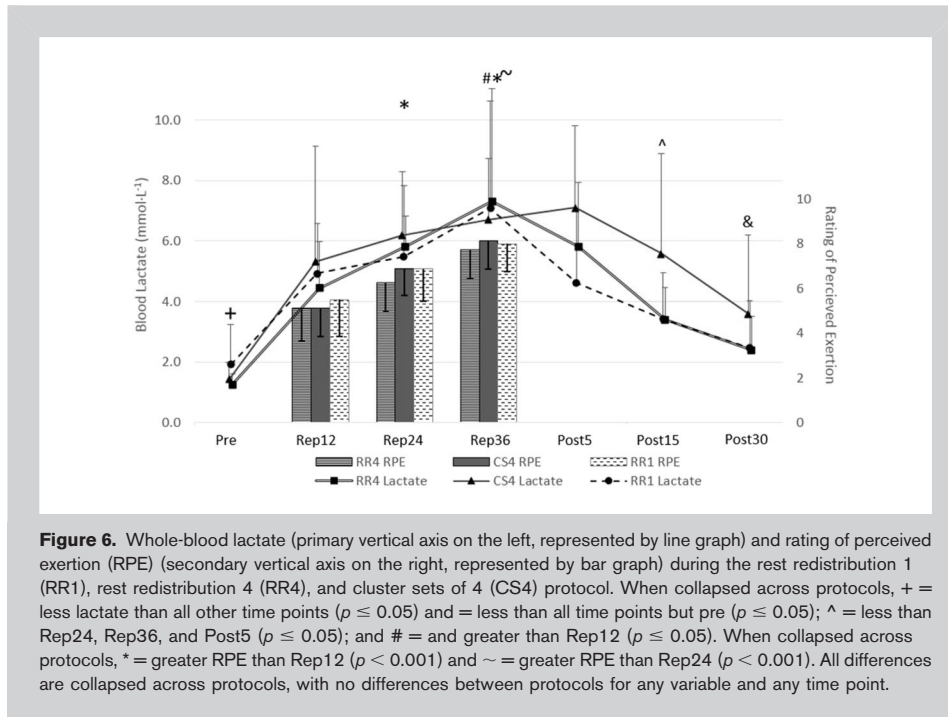
A certified phlebotomist also performed finger stick blood sampling at the following time points: PRE, Rep12, Rep24, Rep36, 5 minutes postexercise (P5), P15, and P30. These samples were used for the analysis of whole blood lactate (La) using a Lactate Pro 2 device (Arkray Global Business, Inc., Kyoto, Japan). The first drop of blood was wiped away using medical gauze and a subsequent drop was used as the blood sample. The device instantaneously analyzed each sample, and the values were recorded for analysis.

Statistical Analyses

Descriptive statistics were calculated for RPE, MVM, MVL, La, C, GH, TT, and SHBG. Separate $3 (\text{protocol}) \times 4 (\text{time})$ repeated measures analysis of variance (ANOVA) were used for C, GH, TT, and SHBG. A $3 (\text{protocol}) \times 7 (\text{time})$ repeated measures ANOVA was used for La. Separate $3 (\text{protocol}) \times 3 (\text{time})$ repeated measures ANOVAs were used for RPE and MVL, whereas a $3 (\text{protocol}) \times 1 (\text{time})$ repeated measures ANOVA was used for MVM. The alpha level was set at $p \leq 0.05$, and all statistical analyses were performed using SPSS 22.0 (IBM, Armonk, NY, USA). In addition to these statistics, the magnitude of effect was assessed to determine practical differences in performance variables due to set structures. Therefore, Hedge's g effect sizes (with 95% confidence intervals) were calculated for performance variables (i.e., MVM and MVL) and were interpreted as small (0.2–0.49), moderate (0.5–0.79), and large (≥ 0.8).

RESULTS

Repeated measures ANOVA tests showed that there were no differences between protocols ($p = 0.733$) for MVM (CS4 = $84.3 \pm 6.8\%$; RR4 = $84.5 \pm 6.6\%$; and RR1 = $85.8 \pm 6.0\%$) and only negligible to small effects were present: RR1-CS4, 0.23 (–0.76 to 1.21); RR1-RR4, 0.19 (–0.79 to 1.17); and RR4-CS4, 0.04 (–0.94 to 1.02). There were significant main effects for protocol ($p = 0.026$) and time ($p = 0.017$) for MVL (Figure 2). Effect sizes indicated that different protocols had different effects on MVL. For RR1-CS4, the effect decreased from large at the beginning of the protocol to moderate at the end, 1.01 (–0.03 to 2.05) to 0.56 (–0.44 to 1.56). For RR1-RR4, the effect also decreased from large at the beginning of the protocol to moderate at the end, 1.00 (–0.04 to 2.04) to 0.38 (–0.61 to 1.36). For RR4-CS4, the effect remained negligible from the beginning of the protocol to



the end, 0.08 (−0.90 to 1.06) to 0.18 (−0.81 to 1.16). For C, there was a main effect for protocol ($p = 0.038$), but not for time ($p = 0.079$) (Figure 3). There was a main effect for time ($p = 0.002$) for GH, but there were no differences between protocols ($p = 0.418$) (Figure 4). Similarly, there was a main effect for time ($p = 0.008$) for TT, but there were no differences between protocols ($p = 0.523$) (Figure 5). For SHBG, there was neither a main effect for protocol ($p = 0.995$) nor time ($p = 0.389$) (Figure 6). For both La and RPE, there was a main effect for time ($p \leq 0.001$), but not for protocol ($p = 0.089$ for La; $p = 0.766$ for RPE) (Figure 6). There were no protocol-time interactions present for any variable ($p > 0.05$; range $p = 0.221$ – 0.917).

DISCUSSION

The purpose of this study was to compare the effects of a basic hypertrophy-oriented CS protocol (44) with 2 different RR protocols on the kinematic, hormonal, metabolic, and perceptual responses of back squats in trained men. The main finding was that when using the same load and number of repetitions, MVM and the perceptual, metabolic, and hormonal responses to back squats were similar regardless of how the rest periods were distributed within the session. However, the effect of RR1 on minimizing MVL in the beginning stages of the protocol was large compared with CS4 and RR4, but decreased to a moderate effect by the end of the protocol. Therefore, these data collectively show that shorter but more frequent rest intervals may spare neuromuscular performance in the beginning stages of a resistance training session, but such an advantage decreases as the

session's duration increases, with all protocols being equally fatiguing and resulting in similar RPE values and metabolic and hormonal changes.

Simultaneous increases in RPE, decreases in movement velocity, and increases in La during resistance training are evident throughout the literature (9,17,38), with the same occurring in this study. As the protocols within this study were designed to avoid RM loading and muscular failure (44), RPE was not expected to reach maximal values despite performing the concentric movements with maximal intent (17). Specifically, RPE increased equally throughout each protocol from 5.3 after the 12th repetition to 6.7 after the 24th repetition, and up to an average of 8.0 out of 10 after

the final repetition, indicating that each protocol could be considered as “hard” (31). Previous research has shown similar RPE values during basic CS protocols inclusive of an interrepetition rest (17). In the study by Hardee et al. (17), subjects performed 3 TS of 6 power cleans using 80% 1RM with 3 minutes of interset rest, resulting in RPE scores of approximately 6.0, 7.5, and 9.0 after each set, respectively. On the other hand, RPE was significantly less during CS, resulting in RPE scores of approximately 5.5, 6.5, and 7.5 when 20 seconds of interrepetition rest was used, and values of 4.0, 5.0, and 6.0 when 40 seconds of interrepetition rest was used. Therefore, the 20 seconds of interrepetition rest protocol used by Hardee et al. (17) resulted in similar RPE values as those seen in this study. Although a direct comparison between these 2 studies is not possible because of different exercises, loads, and number of repetitions, RPE served as an accurate measure of perceived exertion in both studies, evidenced by decreases in power output (17) and MVL in this study. Furthermore, other research (28) has shown that when the work-to-rest ratio of 5 sets of squats using TS with a 10RM load and 3 minutes of interset rest periods is divided to create a protocol similar to that of the present RR1 protocol, RPE values decreased compared with TS and were similar to those of this study. Therefore, as a subjective measure of perceived exertion that is linked to neuromuscular fatigue, RPE has again proven to be a valid tool that is quick and easy to use (31). However, as coaches and researchers may be more interested in the mechanical and physiological responses that coincide with increases in RPE, more objective measures of fatigue could be useful.

The data in Figure 6 show that increases in RPE throughout the exercise sessions were suggestive of increases in La, in agreement with previous research (2,23,26). The La values of this study were greater than (4) and less than (7) those of other studies most likely because of differences in exercise choice, order, and external loading parameters. Nonetheless, a gradual increase in La during exercise and the return of La to near baseline levels 30 minutes after exercise in this study was similar to the response seen in other studies (4,7,12,33). In a study by Oliver et al. (33), subjects completed 4 sets of 10 back squats with 70% 1RM using either a TS (4 × 10 with a 120-second interset rest) or an RR protocol (4 × 10 with a 90-second interset rest and 30-second intraset rest after the fifth repetition of each set). Similar to this study, both protocols resulted in increased blood La during exercise and up to 30 minutes after exercise, but La was greater at all time points in response to the TS protocol (33). Goto et al. (12) showed that La, although not measured during exercise, was greater after a TS protocol using 10RM loads with 1 minute of interset rest during the lat pull-down, shoulder press, and knee extension exercises compared with a basic CS design in which the load and number of repetitions were equalized to the TS protocol. Together, the previous research shows that RR (4,7,33) and basic CS (12) protocols result in less La than TS. In addition, this study shows that basic CS and RR protocol designs result in similar metabolic responses, suggesting that the protocols used in this study may have also resulted in less La compared with a TS protocol, had one been implemented in the study. Although not measured in any of the previously mentioned studies, it is likely that changes in La occurred in response to ATP/PCr availability (9,11,38). Along these lines, changes in energy availability typically affect performance, as evidenced in many studies (4,9,10,37).

Specifically, decreases in ATP/PCr availability and increases in La have been associated with decreases in power output and movement velocity (11,33,38). The same held true during this study, as all protocols resulted in increases in La and decreased movement velocity, indicated by MVL. Despite La being similar between protocols, RR1 had a large effect on MVL compared with RR4 and CS4, indicating that mean velocity decreased less during RR1 in the beginning stages of the session. Although, by the end of the protocol, the effect of RR1 compared with RR4 and CS4 was only moderate. In addition, RR1 experienced less MVL than CS4 when measurements were collapsed across time, indicating that MVL during RR1 was less than CS4. On the other hand, MVM showed that when the mean velocity of all 36 repetitions was taken into consideration, all the protocols maintained mean velocity to a similar extent. Therefore, it is important for the strength and conditioning professional to determine what is most important when designing a resistance training program using different CS configurations: the global response (i.e., MVM and similar measures that account for all repetitions) or the

response at specific time points (i.e., MVL and other measures that are time sensitive). Previous research has shown that mean velocity is better maintained during basic CS structures compared with TS with 2 minutes of interset rest (45) and when RR is compared with TS (30,34). However, this is the first study to show that although basic CS and RR result in similar MVM, redistributing the rest periods of a basic CS protocol can affect MVL at different stages within a resistance training session. Therefore, researchers and practitioners should acknowledge that all CS subtypes (43) may not result in the same acute responses to exercise, bearing in mind that future research should further investigate such effects before a conclusive statement on the topic can be made.

Lastly, the development of hypertrophy is a complex process, during which shifts in the acute hormonal milieu play a role (13,24,27). Research has shown that as La increases, there is usually a concomitant increase in GH (8,22), resulting in associations to be made between La and the anabolic process. Therefore, the initial scientific reviews of CS did not recommend CS for hypertrophy-based workouts, as the additional rest periods were thought to result in less La and a blunted hormonal response compared with TS (14,15). On the contrary, more recent research suggests the opposite (32,35), indicating that when designed appropriately, CS protocols may be just as effective as TS at developing muscle hypertrophy (43).

In support of this, this study showed that CS4, RR1, and RR4 resulted in similar increases in La, increases in post-exercise GH levels, and decreases in postexercise TT levels. Because GH plays a role in tissue regeneration (8,24), equivalent increases in GH after exercise show that all 3 protocols may have resulted in similar anabolic stimuli. Furthermore, an increased GH response paired with no change in post-exercise C levels further shows that the mechanical stress provided during each of the protocols most likely favored anabolism over catabolism. Although decreases in TT can be considered as unwarranted for hypertrophy, there was a significant decrease in TT at P60. Despite a lack of significant changes in SHBG, an apparent increase in SHBG from P30 to P60 combined with a significant decrease in TT (Figure 5) indicates that the transport of TT via SHBG may have increased, resulting in a possible increased uptake of TT into the muscles, which may result in the increase of muscle growth. Therefore, these data indicate that RR and CS protocols can create a physiological environment indicative of skeletal muscle hypertrophy, but other research using CS does not agree. Despite the basic CS protocol used by Goto et al. (12) resulting in significant increases in La, the apparent elevations in postexercise GH were not significant and TT remained unchanged compared with pre-exercise values. The lack of postexercise endocrine response documented in that study (12) infers that the stimulus provided during their protocol, which included cadence-controlled machine-based lat pull-down, shoulder press, and knee extensions,

may not have been sufficient to increase La enough to stimulate a subsequent endocrine response (8,24). However, data from other research agree with data of this study, showing that RR protocols inclusive of maximal velocity compound lower-body movements have resulted in proanabolic shifts in TT (33) and GH (7) concentrations, albeit to a lesser extent than a TS protocol with 2 minutes of interset rest periods (33). The lack of a TS protocol within this study does not allow for a comparison to be made between TS and the CS subtypes used in this study, but the purpose of this study was to investigate the effects of novel CS and RR structures. Still, the hormonal data presented in this study are comparable with data of previous studies using similar hypertrophic protocols of 4 sets of 10 back squats with 70–75% 1RM using TS with 1.5–2 minutes of interset rest (29,33). Therefore, when designed appropriately, basic CS and RR protocols may result in acute hormonal responses comparable with TS protocols designed to stimulate hypertrophy.

PRACTICAL APPLICATIONS

In conclusion, this is the first study to investigate the effects of a basic CS protocol to 2 RR protocols on the hormonal, metabolic, and perceptual responses of back squats in trained men. Collectively, these data show that when using the same load, total rest time, and number of repetitions, the perceptual, metabolic, and hormonal responses to back squats were generally similar regardless of how the rest periods were distributed within the session. Comparable increases in RPE and La between the protocols also support that redistributing rest periods within a resistance training session can result in similar exertion and energetic demands. The large effects of RR1 on MVL indicate that shorter but more frequent rest intervals may spare performance in the beginning stages of a longer protocol, but such an advantage is reduced as the protocol continues: hinting that mechanical changes may occur in the absence of metabolic or hormonal changes. Lastly, this study provides further evidence that different types of CS protocols can be designed in a way to increase the acute hormonal response to exercise, possibly stimulating the anabolic process. The data from this study indicate that coaches can implement shorter but more frequent rest intervals during CS or RR protocols to spare neuromuscular performance in the beginning stages of a resistance training session while still stimulating the anabolic process. Nevertheless, further research must investigate the effectiveness of such protocols within a chronic training environment.

ACKNOWLEDGMENTS

The completion of this study and preparation of this manuscript were partially funded by a postgraduate research scholarship at Edith Cowan University and a Biological Aspects of Human Movement grant (PRVOUK P38) from Charles University. Neither Edith Cowan University nor

Charles University influenced the experimental design, data analysis, or presentation of the study.

REFERENCES

1. Abe, T, Kearns, CF, and Sato, Y. Muscle size and strength are increased following walk training with restricted venous blood flow from the leg muscle, Kaatsu-walk training. *J Appl Physiol* 100: 1460–1466, 2006.
2. Aniceto, RR, Ritti-Dias, RM, Dos Prazeres, TM, Farah, BQ, and de Lima, FF. do Prado, WL. Rating of perceived exertion during circuit weight training: a concurrent validation study. *J Strength Cond Res* 29: 3336–3342, 2015.
3. de Souza, TP Jr, Fleck, SJ, Simão, R, Dubas, JP, Pereira, B, de Brito Pacheco, EM, da Silva, AC, and de Oliveira, PR. Comparison between constant and decreasing rest intervals: Influence on maximal strength and hypertrophy. *J Strength Cond Res* 24: 1843–1850, 2010.
4. Denton, J and Cronin, JB. Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res* 20: 528–534, 2006.
5. Fleck, SJ and Kraemer, WJ. Types of Strength Training. In: 4th, ed. *Designing Resistance Training Programs*. Champaign, IL: Human Kinetics, 2014. pp. 15–62.
6. Folland, JP, Irish, CS, Roberts, JC, Tarr, JE, and Jones, DA. Fatigue is not a necessary stimulus for strength gains during resistance training. *Br J Sports Med* 36: 370–373, 2002.
7. Girman, JC, Jones, MT, Matthews, TD, and Wood, RJ. Acute effects of a cluster-set protocol on hormonal, metabolic and performance measures in resistance trained males. *Eur J Sport Sci* 14: 151–159, 2014.
8. Godfrey, RJ, Madgwick, Z, and Whyte, GP. The exercise-induced growth hormone response in athletes. *Sports Med* 33: 599–613, 2003.
9. Gorostiaga, EM, Navarro-Amezqueta, I, Calbet, JA, Hellsten, Y, Cusso, R, Guerrero, M, Granados, C, Gonzalez-Izal, M, Ibanez, J, and Izquierdo, M. Energy metabolism during repeated sets of leg press exercise leading to failure or not. *PLoS One* 7: e40621, 2012.
10. Gorostiaga, EM, Navarro-Amezqueta, I, Calbet, JA, Sanchez-Medina, L, Cusso, R, Guerrero, M, Granados, C, Gonzalez-Izal, M, Ibanez, J, and Izquierdo, M. Blood ammonia and lactate as markers of muscle metabolites during leg press exercise. *J Strength Cond Res* 28: 2775–2785, 2014.
11. Gorostiaga, EM, Navarro-Amezqueta, I, Cusso, R, Hellsten, Y, Calbet, JA, Guerrero, M, Granados, C, Gonzalez-Izal, M, Ibanez, J, and Izquierdo, M. Anaerobic energy expenditure and mechanical efficiency during exhaustive leg press exercise. *PLoS One* 5: e13486, 2010.
12. Goto, K, Ishii, N, Kizuka, T, and Takamatsu, K. The impact of metabolic stress on hormonal responses and muscular adaptations. *Med Sci Sports Exerc* 37: 955–963, 2005.
13. Gotshalk, LA, Loebel, CC, Nindl, BC, Putukian, M, Sebastianelli, WJ, Newton, RU, Häkkinen, K, and Kraemer, WJ. Hormonal responses of multisets versus single-set heavy-resistance exercise protocols. *Can J Appl Physiol* 22: 244–255, 1997.
14. Haff, G, Burgess, S, and Stone, M. Cluster training: Theoretical and practical applications for the strength and conditioning professional. *Prof Strength Cond* 12: 12–17, 2008.
15. Haff, GG, Hobbs, RT, Haff, EE, Sands, WA, Pierce, KC, and Stone, MH. Cluster training: A novel method for introducing training program variation. *Strength Cond J* 30: 67–76, 2008.
16. Hansen, KT, Cronin, JB, and Newton, MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perf* 6: 455–468, 2011.

17. Hardee, JP, Lawrence, MM, Utter, AC, Triplett, NT, Zwetsloot, KA, and McBride, JM. Effect of inter-repetition rest on ratings of perceived exertion during multiple sets of the power clean. *Eur J Appl Physiol* 112: 3141–3147, 2012.
18. Iglesias-Soler, E, Mayo, X, Rio-Rodriguez, D, Carballeira, E, Farinas, J, and Fernandez-Del-Olmo, M. Inter-repetition rest training and traditional set configuration produce similar strength gains without cortical adaptations. *J Sports Sci* 34:1473–1484, 2016.
19. Izquierdo, M, Ibanez, J, Gonzalez-Badillo, JJ, Häkkinen, K, Ratamess, NA, Kraemer, WJ, French, DN, Eslava, J, Altadill, A, Asiain, X, and Gorostiaga, EM. Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. *J Appl Physiol* 100: 1647–1656, 2006.
20. Joy, JM, Oliver, JM, McCleary, SA, Lowery, RP, and Wilson, JM. Power output and electromyography activity of the back squat exercise with cluster sets. *J Sports Sci* 1: 37–45, 2013.
21. Kraemer, WJ, Gordon, S, Fleck, S, Marchitelli, L, Mello, R, Dziados, J, Friedl, K, Harman, E, Maresch, C, and Fry, A. Endogenous anabolic hormonal and growth factor responses to heavy resistance exercise in males and females. *Int J Sports Med* 12: 228–235, 1991.
22. Kraemer, WJ, Marchitelli, L, Gordon, SE, Harman, E, Dziados, JE, Mello, R, Frykman, P, McCurry, D, and Fleck, SJ. Hormonal and growth factor responses to heavy resistance exercise protocols. *J Appl Physiol* 69: 1442–1450, 1990.
23. Kraemer, WJ, Noble, BJ, Clark, MJ, and Culver, BW. Physiologic responses to heavy-resistance exercise with very short rest periods. *Int J Sports Med* 8: 247–252, 1987.
24. Kraemer, WJ and Ratamess, NA. Hormonal responses and adaptations to resistance exercise and training. *Sports Med* 35: 339–361, 2005.
25. Krieger, JW. Single vs. multiple sets of resistance exercise for muscle hypertrophy: A meta-analysis. *J Strength Cond Res* 24: 1150–1159, 2010.
26. Lagally, KM, Robertson, RJ, Gallagher, KI, Goss, FL, Jakicic, JM, Lephart, SM, McCaw, ST, and Goodpaster, B. Perceived exertion, electromyography, and blood lactate during acute bouts of resistance exercise. *Med Sci Sports Exerc* 34: 552–559, 2002.
27. Linnamo, V, Pakarinen, A, Komi, PV, Kraemer, WJ, and Häkkinen, K. Acute hormonal responses to submaximal and maximal heavy resistance and explosive exercises in men and women. *J Strength Cond Res* 19: 566–571, 2005.
28. Mayo, X, Iglesias-Soler, E, and Fernández-Del-Olmo, M. Effects of set configuration of resistance exercise on perceived exertion. *Percept Mot Skills* 119: 825–837, 2014.
29. McCaulley, GO, McBride, JM, Cormie, P, Hudson, MB, Nuzzo, JL, Quindry, JC, and Travis Triplett, N. Acute hormonal and neuromuscular responses to hypertrophy, strength and power type resistance exercise. *Eur J Appl Physiol* 105: 695–704, 2009.
30. Moreno, SD, Brown, LE, Coburn, JW, and Judelson, DA. Effect of cluster sets on plyometric jump power. *J Strength Cond Res* 28: 2424–2428, 2014.
31. Morishita, S, Yamauchi, S, Fujisawa, C, and Domen, K. Rating of perceived exertion for quantification of the intensity of resistance exercise. *Int J Phys Med Rehab* 1: 172, 2014.
32. Oliver, JM, Jagim, AR, Sanchez, AC, Mardock, MA, Kelly, KA, Meredith, HJ, Smith, GL, Greenwood, M, Parker, JL, Riechman, SE, Fluckey, JD, Crouse, SF, and Kreider, RB. Greater gains in strength and power with intraset rest intervals in hypertrophic training. *J Strength Cond Res* 27: 3116–3131, 2013.
33. Oliver, JM, Kreutzer, A, Jenke, S, Phillips, MD, Mitchell, JB, and Jones, MT. Acute response to cluster sets in trained and untrained men. *Eur J Appl Physiol* 115: 2383–2393, 2015.
34. Oliver, JM, Kreutzer, A, Jenke, SC, Phillips, MD, Mitchell, JB, and Jones, MT. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res* 30: 235–243, 2016.
35. Pareja-Blanco, F, Rodriguez-Rosell, D, Sanchez-Medina, L, Sanchis-Moysi, J, Dorado, C, Mora-Custodio, R, Yanez-Garcia, JM, Morales-Alamo, D, Perez-Suarez, I, Calbet, JA, and Gonzalez-Badillo, JJ. Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations. *Scand J Med Sci Sports* 27: 724–735, 2017.
36. Raastad, T, Bjørø, T, and Hallen, J. Hormonal responses to high-and moderate-intensity strength exercise. *Eur J Appl Physiol* 82: 121–128, 2000.
37. Sahlin, K and Ren, JM. Relationship of contraction capacity to metabolic changes during recovery from a fatiguing contraction. *J Appl Physiol* 67: 648–654, 1989.
38. Sanchez-Medina, L and Gonzalez-Badillo, JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc* 43: 1725–1734, 2011.
39. Schoenfeld, BJ. The mechanisms of muscle hypertrophy and their application to resistance training. *J Strength Cond Res* 24: 2857–2872, 2010.
40. Schoenfeld, BJ. Potential mechanisms for a role of metabolic stress in hypertrophic adaptations to resistance training. *Sports Med* 43: 179–194, 2013.
41. Selye, H. Stress and the general adaptation syndrome. *Br Med J* 1: 1383, 1950.
42. Takada, S, Okita, K, Suga, T, Omokawa, M, Kadoguchi, T, Sato, T, Takahashi, M, Yokota, T, Hirabayashi, K, and Morita, N. Low-intensity exercise can increase muscle mass and strength proportionally to enhanced metabolic stress under ischemic conditions. *J Appl Physiol* 113: 199–205, 2012.
43. Tufano, JJ, Brown, LE, and Haff, GG. Theoretical and practical aspects of different cluster set structures: A systematic review. *J Strength Cond Res* 31: 848–867, 2017.
44. Tufano, JJ, Conlon, JA, Nimphius, S, Brown, LE, Banyard, HG, Williamson, BD, Bishop, LG, Hopper, AJ, and Haff, GG. Cluster sets permit greater mechanical stress without decreasing relative velocity. *Int J Sports Physiol Perf* 2016. Epub ahead of print.
45. Tufano, JJ, Conlon, JA, Nimphius, S, Brown, LE, Seitz, LB, Williamson, BD, and Haff, GG. Maintenance of velocity and power with cluster sets during high-volume back squats. *Int J Sports Physiol Perf* 11: 885–892, 2016.
46. West, DW and Phillips, SM. Anabolic processes in human skeletal muscle: Restoring the identities of growth hormone and testosterone. *Phys Sports Med* 38: 97–104, 2010.
47. Yasuda, T, Ogasawara, R, Sakamaki, M, Bemben, MG, and Abe, T. Relationship between limb and trunk muscle hypertrophy following high-intensity resistance training and blood flow-restricted low-intensity resistance training. *Clin Physiol Funct Imaging* 31: 347–351, 2011.